

Assignment Project Work

Hochschule Karlsruhe
University of
Applied Sciences
Fakultät für
Maschinenbau und
Mechatronik



Project Title:

Design and Prototype Implementation of a System Architecture for Automated Data Integration for Specialized Visualization Tools.

Background on the Project Topic:

This project is being conducted in collaboration with the University of Oviedo. The partner university is actively engaged in data-driven research areas where enormous amounts of complex information must be processed. Oncology and genetics serve as the central use cases here: The analysis of DNA sequences generates data streams that can hardly be handled efficiently with traditional tools.

To support the exploratory analysis of such data, an interactive visualization tool was developed at the partner university. For this tool to be used effectively, a powerful system environment is required that automatically acquires the necessary research data, filters it, and converts it into a usable structure.

Project Objective:

The goal of the project is to develop a universally applicable interface solution for automated data acquisition. The medical research question posed by the University of Oviedo serves as a validating use case.

The task is to design an application component that identifies and retrieves relevant datasets from public repositories based on defined parameters and prepares them for the visualization layer. Particular emphasis is placed on the flexibility of the system to handle evolving data structures and conceptual relationships (ontologies) within the application domain. The choice of technological tools will be made by the students during the design phase..

The project work will include the following objectives and deliverables:

- Familiarization with the state of the art (e.g., ontologies, databases, and their programming languages)
- Concept and system design
- Implementation
- Validation of the system
- Evaluation and opportunities for improvement
- Documentation and presentation

B Offered in:
SS 2026

Target audience:
Bachelor's/Master's/degree program

Project code:
26ss_NN_Kurztitel

Advisor (Professor): Prof. Catherina Burghart; E-Mail: catherina.burghart@h-ka.de
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Signature and date (staff member and supervisor)